

 timeisbrain	CLINICAL INVESTIGATION PLAN PROMISE20 RELOADED	CODE: PROMISE20 RELOADED CIP REV: 1 DATE: 11/04/2025 PAGE: 1 of 44
--	---	--

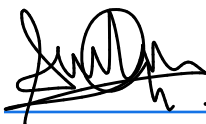
PROMISE20 RELOADED

Somatosensory Evoked Potentials (SEP) Monitoring in Patients with Acute Ischemic Stroke (AIS) and Anterior Large Vessel Occlusion (LVO) Undergoing Endovascular Thrombectomy (EVT). A Clinical Validation of the Updated BrainN20® Medical Device.

Sponsor: **Time is Brain, S.L.**

Carretera de Canyet s/n

08916 Badalona, Barcelona, Spain

Code:	PROMISE20 RELOADED
Version:	1
Date of version:	11/04/2025
Author approval:	Prof. Antoni Dávalos, MD, PhD CSO-CMO Time is Brain S.L.  Antoni Dávalos (Apr 10, 2025 17:01 GMT+2) <i>The signature on this page indicates review and approval of the final version of the protocol.</i>

REVISION HISTORY		
Revision N°	Edition date	Reason
1		First release

 timeisbrain	CLINICAL INVESTIGATION PLAN PROMISE20 RELOADED	CODE: PROMISE20 RELOADED CIP REV: 1 DATE: 11/04/2025 PAGE: 2 of 44
--	---	--

SPONSOR AND SITE PRINCIPAL INVESTIGATOR AGREEMENT ON THE PROTOCOL

STUDY CODE: PROMISE20 RELOADED

Study Title: Somatosensory Evoked Potentials (SEP) Monitoring in Patients with Acute Ischemic Stroke (AIS) and Anterior Large Vessel Occlusion (LVO) Undergoing Endovascular Thrombectomy (EVT). A Clinical Validation of the Updated Brain20® Medical Device. PROMISE20 RELOADED

Sponsor: Time is Brain, S.L.

Carretera de Canyet s/n

08916 Badalona, Barcelona, Spain

That the clinical study respects the ethical regulations applicable to this type of study.

That agree to participate as principal investigator in this clinical study.

That has the necessary material and human resources to carry out the clinical study, without interfering with the performance of other types of studies or other tasks usually entrusted to it.

That it is committed that each subject is treated and controlled following what is established in the protocol with a favourable opinion by the Clinical Investigation Ethics Committee and authorized by the Agencia Española de Medicamentos y Productos Sanitarios (Spanish competent authority).

That it will respect the ethical and legal standards applicable to this type of study and will follow the standards of good clinical practice in its conduct.

That the collaborators you need to carry out the proposed clinical study are suitable.

By signing this document, I confirm that the clinical study will be conducted in accordance with the protocol and all applicable regulations.

Principal investigators undertake to obtain signed commitments from their collaborators.

Name/function or role: Alicia Martínez / Time is Brain, S.L. Legal Representative and CEO. Date:	Name/function or role: Marc Ribó / IP. Site: H.U. Vall d'Hebron, Barcelona, Spain Date:
Name/function or role: Pol Camps / IP. Site: Hospital Santa Creu i Sant Pau, Barcelona, Spain. Date:	Name/function or role: Pere Cardona / IP. Site: H. U. Bellvitge, Hospitalet (Barcelona), Spain. Date:

1. INDEX

1. INDEX	3
2. LIST OF ABBREVIATIONS	5
3. GENERAL.....	6
3.1. Scope	6
3.2. Regulatory basis	6
3.3. Protocol synopsis	6
4. IDENTIFICATION AND DESCRIPTION OF THE INVESTIGATIONAL DEVICE	10
5. JUSTIFICATION FOR THE DESIGN OF THE CLINICAL INVESTIGATION.....	13
5.1. Background and rationale	13
5.2. Trial design	14
5.3. Clinical development plan	15
6. BENEFITS AND RISKS OF THE INVESTIGATIONAL DEVICE, CLINICAL PROCEDURE, AND CLINICAL INVESTIGATION	15
6.1. Anticipated clinical benefits	15
6.2. Anticipated adverse-device effects.....	17
6.3. Risks associated with the participation in the clinical investigation.....	20
6.4. Possible interaction with concomitant medical treatments considered under the risk analysis	20
6.5. Steps that will be taken to control or mitigate the risks	20
6.6. Rationale for benefit-risk ratio.....	20
7. OBJECTIVES AND HYPOTHESES OF THE CLINICAL INVESTIGATION	21
7.1. Hypothesis.....	21
7.2. Objectives	21
8. DESIGN OF THE CLINICAL INVESTIGATION.....	22
8.1. Study setting.....	22
8.2. Subjects	22
8.2.1. Inclusion Criteria	22
8.2.2. Exclusion Criteria.....	23
8.2.3. Enrolment: Patients will be enrolled in a single arm and consecutively	23
8.2.4. Duration of subject's participation.....	23
8.2.5. Subject replacement	23
8.2.6. Lost to follow-up	24
8.3. Endpoints and outcome.....	24
8.4. Blinding.....	25
8.5. Variables	25
8.5.1. At Baseline before EVT	25
8.5.2. During the EVT procedure	25
8.5.3. During the ASU admission	26
8.5.4. After discharge	26
8.5.5. Times.....	26
8.5.6. Participant timeline	27
8.6. Investigational device	27
8.7. Procedures	28
8.8. Data monitoring.....	29



timeisbrain

CLINICAL INVESTIGATION PLAN PROMISE20 RELOADED

CODE: **PROMISE20 RELOADED CIP**
REV: 1
DATE: 11/04/2025
PAGE: 4 of 44

9. STATISTICAL DESIGN AND ANALYSIS	30
9.1. Sample size.....	30
9.2. Statistical considerations	30
9.3. Analysis Populations	30
9.4. General statistical approach.....	30
9.5. Primary Analyses	30
Secondary Analyses	31
9.6. 31	
9.7. Subgroups analysis	31
9.8. Multiple comparison / Multiplicity	31
9.9. Pooling Strategy	31
9.10. Missing data.....	31
9.11. Variables transformation	31
9.12. Interim analysis	32
9.13. Independent committees / labs	32
10. DATA MANAGEMENT.....	32
10.1. Data collection methods	32
11. AMENDMENTS TO THE CIP	33
12. DEVIATIONS FROM CLINICAL INVESTIGATION PLAN.....	33
13. DEVICE ACCOUNTABILITY	33
14. STATEMENTS OF COMPLIANCE.....	33
14.1. Initiation of the study	33
14.2. Ethical principles.....	34
14.3. Study financing	34
14.4. Contracts	34
14.5. Insurance	34
14.6. Data protection	34
15. INFORMED CONSENT PROCESS.....	36
16. ADVERSE EVENTS, ADVERSE DEVICE EFFECTS, AND DEVICE DEFICIENCIES	37
17. VULNERABLE POPULATION.....	39
18. SUSPENSION OR PREMATURE TERMINATION OF THE CLINICAL INVESTIGATION	39
19. PUPICATION POLICY	40
20. TRAINING OF INVESTIGATOR AND SITE PERSONNEL	41
21. BIBLIOGRAPHY	41
22. ANNEXES	44

2. LIST OF ABBREVIATIONS

Abbreviation	Definition
AEMPS	Agencia Española de Medicamentos y Productos Sanitarios
AE	Adverse Event
AIS	Acute Ischemic Stroke
ASPECTS	Alberta Stroke Program Early CT
BP	Blood Pressure
CEIm	Ethics Committee (Spain)
CICAT	Codi Ictus Catalunya registry
CIP	Clinical Investigation Plan
CRF	Case Report Form
CT	Computerized Tomography
CTA	Computerized Axial Tomography Angiography
DIC	Deferred Informed Consent
DSMB	Data Safety Monitoring Board
DD	Device Deficiency
DWI	Diffusion-Weighted Imaging
eCRF	Electronic Case Report Form
EC	Ethics Committee
ESO	European Stroke Organization
EVT	Endovascular Thrombectomy
IB	Investigator's Brochure
ICH	Intracerebral Hemorrhage
ICF	Informed Consent Form
IRB	Institutional Review Board
MCA	Middle Cerebral Artery
MRI	Magnetic Resonance Image
MR	Magnetic resonance
MRA	Magnetic resonance angiography
mRS	Modified Rankin Scale
eTICI	Expanded Treatment In Cerebral Infarction scale
M1	Proximal segment of the MCA from the origin to bifurcation/trifurcation, also known as horizontal or sphenoidal segment
M2	Distal segment of the MCA, also known as insular segment, from bi(tri)furcation to circular sulcus of insula
NCCT	Non-contrast computed tomography
NIHSS	National Institute of Health Stroke Scale
RACE	Rapid Arterial occlusion Evaluation
SAE	Serious Adverse Event
TNK	Tecnetepase

3. GENERAL

3.1. Scope

This document is the clinical investigation plan (CIP) for the conduct of a pre-market pivotal clinical trial with the medical device (MD) BrainN20®, to be submitted to the ethics committee of H.U. Vall d'Hebron, Barcelona, Spain. Reference EC and to the Spanish Competent Authority (Agencia Española de Medicamentos y Productos Sanitarios, AEMPS), for review and authorisation of the investigation.

3.2. Regulatory basis

This CIP has been prepared in accordance with the regulation listed below:

- *Regulation EU/2017/745 of 5 April 2017 on medical devices, as amended (MDR, Articles 62 to 80 and Annex XV),*
- *MDR transposition into Spanish legislation in Real Decreto 192/2023 of March 21 which regulates medical devices, the Royal Decree 1090/2015, 4 December, regulating the clinical trials with drugs, Ethics Committees and the Spanish Register of Clinical Studies,*
- *and with the principles of good clinical practice as described in Annex A of standard ISO 14155:2020 – Clinical investigation of medical devices for human subjects — Good clinical practice.*

3.3. Protocol synopsis

Title	Somatosensory Evoked Potentials (SEP) Monitoring in Patients with Acute Ischemic Stroke (AIS) and Anterior Large Vessel Occlusion (LVO) Undergoing Endovascular Thrombectomy (EVT). A Clinical Validation of the Updated BrainN20® Medical Device.
Background	The PROMISE trial (Stroke: Vascular and Interventional Neurology 2023;3:e000735) investigated the presence and amplitude of the N20 response of Somatosensory Evoked Potentials (SEP) recorded before endovascular thrombectomy (EVT) in the ischemic hemisphere of 223 patients with acute ischemic stroke (AIS) and anterior large vessel occlusion (LVO). Presence of N20 predicted functional independence at day 7 with a sensitivity of 93% (95% CI, 78-98%) and negative predictive value of 93% (80-98%), which increased to 100% after the EVT. Its predictive performance before treatment was superior to that of common factors used in clinical routine and added 0.07 points to the predictive model combining clinical variables, NIHSS score and ASPECTS (AUC of 0.84 versus AUC 0.77). Furthermore, N20+ had a higher predictive value of good outcome than advanced imaging variables recorded at baseline, such as the volume of infarct core, volume of hypoperfusion and collateral circulation. Taken together, these findings indicate that N20 might be a neurophysiological surrogate of salvageable brain and highlight the concept that ischemic brain tissue may remain viable from the perspective of cortical electrical function. Therefore, N20 SEP monitoring might be a novel neurophysiological biomarker in acute stroke with the potential of having a major impact on the way stroke patients are managed throughout the world. Time is Brain S.L. has developed a portable medical device, BrainN20®, which allows a fast (like ECG) non-invasive, real time and automated continuous monitoring of N20 SEP that does not require special training. PROMISE20 was the first in human study (NCT06149754) aiming to confirm the high reliability of the medical device BrainN20® automatic recording of N20. The trial was halted temporally in July 2024 after a futility analysis of the first 17 patients studied due to usability difficulties of the device and electrodes. Tolerability was good and there were no side effects.

 timeisbrain	CLINICAL INVESTIGATION PLAN PROMISE20 RELOADED	CODE: PROMISE20 RELOADED CIP REV: 1 DATE: 11/04/2025 PAGE: 7 of 44
--	---	--

	Technical and usability improvements of BrainN20® device have been implemented and the updated device will be available for use by May 2025. This new trial, PROMISE20 RELOADED, aims to confirm the capability of the updated BrainN20® medical device to accurately monitor the N20 SEP in AIS with LVO undergoing EVT.
Study Design	Prospective, multi-center, open, single arm, interventional study with blinded evaluation of the principal secondary endpoint.
Study Phase	Cohort interventional study.
Clinical Site Locations	Barcelona, Spain
Study Centers	3 Comprehensive Stroke Centers available 24h/7 days
Study Objective	To investigate the reliability of the updated BrainN20® monitoring device to give an accurate diagnostic result of present N20 SEP (N20+), absent N20 SEP (N20-) or non-assessable result ("noisy" N20) according to the classification by two expert physicians blind to the BrainN20® reading results.
Subject Population	Subjects presenting with AIS within 24 hours from symptom onset and anterior LVO undergoing EVT according to local protocols.
Enrollment	Single arm and consecutive. The system is non-invasive and has no safety concerns.
Care providers	Trained vascular neurologists, interventional neuroradiologists and nurses in certified comprehensive stroke centers that treat more than 500 acute stroke patients and perform more than 150 acute mechanical thrombectomies every year.
Interventions	There are no other specific study interventions. If general anesthesia is required, propofol and remifentanyl will be used. Patients will be admitted at acute stroke units (or ICU if needed) and treated following the ESO guidelines.
Diagnostic test	BrainN20® is a portable device ("tablet-shaped") that measures automatically N20 response and characteristics in SEP by using non-invasive surface electrodes on the scalp and wrist. The device provides three diagnostic options: N20 present, N20 absent or N20 non-assessable (noisy), the last in cases where signals have no quality enough to be automatically analyzed, due to radiofrequency or movement artifacts, technical or usability problems. Furthermore, an algorithm, which uses N20 response, provides the probability of having favorable outcome after 7 and 90 days from EVT (i.e. modified Rankin Scale score 0-2). A single BrainN20® monitoring will be performed before EVT and repeated after EVT as soon as possible, within 30 minutes in the angio-room.
Inclusion Criteria	1. Acute ischemic stroke with LVO undergoing treatment with EVT according to local protocols. 2. Age ≥18. 3. No significant pre-stroke functional dependence (mRS ≤ 2). 4. Patient treatable within 24 hours of symptom onset. Symptoms onset is defined as the time the patient was last seen well, and treatment point as the time of arterial puncture. 5. Baseline NIHSS score obtained prior to procedure must be equal or higher than 1 point. 6. Occlusion (TICI 0-1) of the intracranial ICA (distal ICA or T occlusions), MCA-M1 segment or M2 suitable for EVT, as evidenced by CTA or MRA with or without concomitant cervical carotid occlusion or stenosis. 7. Any infarct core, without volume restrictions, measured by ASPECT score on non-contrast CT, or DWI-MRI prior to EVT. 8. Informed consent obtained from patient or acceptable patient surrogate; or the deferred informed consent, to avoid the delay in the start of the mechanical thrombectomy. 9. Because this trial does not provide benefit and does not interact in treatment decision making during the trial, included patients may also participate in other clinical trials.



timeisbrain

CLINICAL INVESTIGATION PLAN

PROMISE20 RELOADED

CODE: **PROMISE20 RELOADED CIP**

REV: 1

DATE: 11/04/2025

PAGE: 8 of 44

Exclusion Criteria	<u>Clinical criteria</u> 1. Patients with a well-documented history of neuromuscular diseases and other neurodegenerative disorders, prior stroke (TIA not excluded) or brain surgery that could interfere with SEP assessment. 2. Serious, advanced, or terminal illness with anticipated life expectancy of less than three months. 3. Patients with a pre-existing neurological or psychiatric disease that would confound the neurological or functional evaluations. 4. Women of childbearing age without a negative pregnancy test <u>Neuroimaging criteria</u> 5. CT or MR evidence of hemorrhage (the presence of microbleeds is allowed). 6. Subjects with occlusions in multiple vascular territories (e.g., bilateral anterior circulation, or anterior/posterior circulation). 7. Evidence of intracranial tumor (except small meningioma). <u>BraiN20® medical device safety issues:</u> 8. Subjects with a demand-type cardiac pacemaker, defibrillator, or other electrical implant or metal. 9. Patients with suspected or well-known cancerous skin lesion in the area where electrical stimulation will be applied. 10. Patients who have a localized disorder in the wrist and forearm where electrical stimulation is to be applied (i.e., fractures or dislocations).
Cohort groups	Modified Rankin Scale score will be measured by blinded local raters according to a structured face to face or telephone interview. Three outcome groups are defined: Stroke good functional recovery: mRS lower or equal than 2 at 90 days. Stroke mild functional recovery: mRS lower or equal than 3 at 90 days. Stroke severe functional dependence or death: mRS equal or higher than 4 at 90 days.
Primary objective	Reliability of the BraiN20® Medical Device automatic result of N20 SEP before EVT will be analysed by concordance with the classification of the monitored SEP recording by two expert physicians blind to BraiN20® reading results and clinical outcome. Three categories are established: present N20 SEP (N20+), absent N20 SEP (N20-) or non-assessable N20 (noisy).
Secondary objectives	The principal secondary objective is to determine the predictive performance of functional outcome at day 90 of an algorithm based on the presence and characteristics of the N20 response before EVT recorded through the BraiN20® device. Other secondary objectives are: • To evaluate whether an interaction exists between the presence of N20 and recanalization in relation to the patient's outcome, as measured by the mRS. • To study if BraiN20® provides at least a proportion of assessable patients (N20+ or N20-) equal or higher than 0.65 for a true proportion of 80%. • To determine the rate of present N20 before EVT by clinical stroke severity (NIHSS score <17 or ≥17), time from symptom onset to first recording (<6h or ≥6h) and ASPECT score (<5, ≥5). • To determine the rate of present N20 before EVT by dramatic early favourable response as determined by an NIHSS of 0-2 or NIHSS improvement ≥ 8 points at 24 (-/+ 6 hours) hours. • To evaluate the predictive performance of the BraiN20® algorithm before EVT on functional outcome at day 7.

	- To estimate the frequency of patients in whom present N20 SEP (N20+) recording prior to treatment has disappeared after the thrombectomy procedure, and the potential associated adverse events. - To estimate the frequency of patients in whom absent N20 SEP (N20-) recording prior to treatment appeared after the thrombectomy procedure - To evaluate the predictive performance of the BrainN20® algorithm after EVT on functional outcome at day 7 and 90 - To describe usability problems - To register the number of consumables used - To list Frequency and causes of non-evaluable recordings of N20. Safety outcomes related to BrainN20® monitoring are: - Mortality at day 7. - Tolerability of the BrainN20® monitoring. - Device deficiency recordings - Serious adverse events (related and not related to the monitoring procedure) within the study period. The time consumption in setting electrodes and monitoring before treatment. - Door to puncture time in patients treated with EVT
Study Timeline (Projected)	Project timelines: - Final Protocol: April 2025. - First Site Activated: June 2025. - First Subject In: June 2025. - Last Subject In: March 2026. - Evaluation of primary end point: June 2026 - Last Subject Out: June 2026. - Database Lock: July 2026. Final Clinical Study Report: October 2026.
Sample size	The proportion of patients with accomplished baseline BrainN20® recording is estimated in 80%. Following previous results, we estimate a real rate of present N20 of 49%, absent N20 of 26% and non-assessable N20 SEP of 25% in acute stroke patients with LVO before EVT. Accordingly, assuming a real kappa (K) value equal to 0.75, in order to prove an agreement of $K > 0.60$ using the lower limit of the 95%CI, 72 patients will be needed. No interim analysis will be done.
Study variables	A specific e-CRF will be generated with a minimum of variables of interest commonly recorded in clinical routine, so they will not require additional work. Deferred Informed Consent (DIC) will be allowed
Statistical analysis	Agreement will be quantified by the Kappa (K) statistic of each examiner with the BrainN20® results (3 categories: N20+, N20- and noisy) and 95%CI will be calculated. In addition, interrater agreement will be estimated. The proportion of good functional recovery will be calculated for both groups (N20+ and N20-). The predictive performance of present N20 SEP before EVT as a biomarker of good functional recovery will be analysed by logistic regression and quantified as odds ratio with 95% confidence interval compared to patients with the presence of N20 SEP. The predictive effect of N20 on good functional recovery, mild recovery and stroke severe dependence or death will be further measured by the metrics of sensitivity, specificity, positive predictive value, and negative predictive value with 95% CI.

 timeisbrain	CLINICAL INVESTIGATION PLAN PROMISE20 RELOADED	CODE: PROMISE20 RELOADED CIP REV: 1 DATE: 11/04/2025 PAGE: 10 of 44
--	---	---

Steering Committee	Chair: Antoni Dávalos, Neurologist, CMO-CSO, Time is Brain S.L. Members: Alicia Martínez-Piñeiro, Department of Neurosciences, Hospital Germans Trias; CEO, Time is Brain S.L., Institut d'Investigació Germans Trias, Badalona Marc Ribó, Department of Neurology, H.U. Vall d'Hebron, Barcelona, Spain. Pol Camps, H. U. Santa Creu i Sant Pau, Barcelona, Spain. Pere Cardona, H. U. Bellvitge, Hospitalet (Barcelona), Spain Jordi Cortés, Department of Statistics and Operations Research, Universitat Politècnica de Catalunya (UPC), Barcelona (Study statistician)
Study Statistician	SAIL S.L.
Study Statistician by TIB	Jordi Cortés Independent investigator (Data Processor)
Sponsor	Time is Brain, S.L. (Data Controller)
Monitoring and Data management team	SAIL S.L.
End point adjudication committee (EAC) SEP evaluation	Roberto Pelaez, Department of Neurosciences, Hospital Germans Trias; Barcelona Enrica Fava (Carlo Besta hospital, Milano)
Participating Comprehensive Stroke Centers	1. H.U. Vall d'Hebron, Barcelona, Spain. Reference EC. 2. H. U. Santa Creu i Sant Pau, Barcelona, Spain. 3. H. U. Bellvitge, Hospitalet (Barcelona), Spain.

4. IDENTIFICATION AND DESCRIPTION OF THE INVESTIGATIONAL DEVICE

Device name / trade name:

Model and/or Catalogue number / name:	BraiN20 / 101.001
Description:	BraiN20® Device for monitoring system.
Trade Name:	timeisbrain.
Trademark:	BraiN20®. Patent and trademark protected.
Product family:	BraiNXX®.

Device general description:

BraiN20® is a medical device system, active, with rechargeable batteries, integrated screen as a user interface, 4G and Bluetooth connected, embedded software, portable, non-invasive and limited skin contact for N20 SEP (somatosensory evoked potential) monitoring.

The objective of the BraiN20® system is to generate and identify the presence or absence of the N20 somatosensory evoked potential (SEP) generated (and in case of detection, its amplitude and latency), so that through the processing of the registration data information can be offered that will be presented on the screen to the system user. In the future, it is expected that BraiN20® can be used for the triage of patients

affected by acute ischemic stroke (AIS) from the moment of symptom onset (both in a pre- and in-hospital environment) and to monitor electrical activity and brain viability throughout the care process, including endovascular treatment (EVT) and after EVT treatment, while AIS patients are admitted in the Stroke Unit or the Intensive Care Unit (ICU), providing continuous and real-time information on the patient's prognosis. Research on potential applications in different acute stroke subtypes are underway. Research on subarachnoid haemorrhage and coma will be planned in the near future.

The BraiN20[®] system is a portable system capable of stimulating and recording at the same time and in a synchronized manner to process the data and robustly analyze the N20 SEP.

BraiN20[®] consists of the main system, charging base, CMAP stimulation and recording electrodes , cables for CMAP stimulation and recording electrodes and acquisition (or recording) electrodes embedded in a textile headband.

The BraiN20[®] device allows you to generate electrical pulses, carry out the acquisition, configure and control the analysis of the recorded signal and display the results on the screen. It is mainly composed of the stimulation module and the acquisition module, which are physically isolated and also have independent battery power systems.

The BraiN20[®] device has a dual-core processor in the acquisition module that allows one of them to be dedicated to acquisition management, interface, storage, communication with the acquisition module and external systems, etc. and the other core exclusively for the processing and detection algorithm of the N20 SEP. In the stimulation module there is a second processor dedicated to stimulation control.

BraiN20[®] analyses EEG and EMG data in real time on the device itself. In addition, it has an SD card for data storage. For the acquisition has standardized DIN 1.5mm type connectors and for stimulation it has the MINI-MED connector suitable for clinical applications (to which the specific cable for the type of electrode must be connected).

Finally, the BraiN20[®] device has an embedded touch screen that allows you to configure and launch both stimulation and acquisition, and subsequently display the N20 SEP amplitude and latency information, and a graph of the N20 SEP measured.

For additional information, see Investigational Brochure-IB in the attachments section below *1*.



timeisbrain

CLINICAL INVESTIGATION PLAN PROMISE20 RELOADED

CODE: **PROMISE20 RELOADED CIP**
REV: 1
DATE: 11/04/2025
PAGE: 12 of 44



5. JUSTIFICATION FOR THE DESIGN OF THE CLINICAL INVESTIGATION

According to the current clinical evaluation (Rev. 4 of 25/Feb/2025) and the post-market clinical follow-up plan-PMCFP (Rev 1 of 25/Feb/2025). See documents in the attachment section below 7, 8, respectively.

5.1. Background and rationale

Endovascular thrombectomy (EVT) is the gold standard treatment in patients with acute ischemic stroke (AIS) and large vessel occlusion (LVO).¹ However, currently, the probability of severe functional dependence or death at 3 months remains between 14-31% according to the results of the five large clinical trials that demonstrated efficacy and safety of EVT.² There are two fundamental problems that need to be overcome to bring this treatment to more patients. First, patients need to seek help more often and sooner after their symptoms start. Second, once the alert occurs, the patients who may benefit from this treatment need to be identified and triaged to the appropriate facility. Unmet needs are the development of a portable, accurate, user-friendly device to quickly determine in real-time along the stroke journey if there is salvageable brain. This would be especially useful at prehospital levels to accelerate the transfer to comprehensive stroke centers and in the angio-room to monitor potential complications of the EVT.

Biomarkers could be useful to determine a priori patients who could most likely benefit from treatment from those in whom EVT would probably be futile. Multiple predictive factors of clinical response have been described.³ Clinical variables include age, baseline National Institute of Health Stroke Scale (NIHSS) score, systolic blood pressure, and hyperglycemia.³⁻⁷ Relevant radiological parameters are early ischemic signs in non-contrast CT (NCCT) assessed by the Alberta Stroke Program Early CT Score (ASPECTS),^{3,8} thrombus extension and location,⁹ collateral status on CT or /MR angiography¹⁰ and perfusion parameters such as infarct core and penumbra volume.¹¹ Most of these factors require evaluation in a stroke center and are not available at previous levels of care, such as the pre-hospital setting or local hospitals.

Recent clinical trials and metaanalysis have demonstrated the benefit of EVT in extended population with time windows up to 24 hours¹²⁻¹³ and large baseline infarct volume.¹⁴ Patients with large ischemic penumbra on perfusion imaging have a higher probability of response¹¹ but EVT is also beneficial in patients evaluated by non-contrast CT and CTA.¹⁵⁻¹⁹ However, those with very low ASPECT scores and longer treatment windows have a high risk of intracerebral hemorrhage (ICH) and only a small likelihood or no benefit from EVT.^{20,21} Current trial data seem too scarce to justify forgoing advanced stroke imaging in extended time windows and future research is required to better identify patients likely to respond to EVT to further optimize the stroke triage process and health care resource utilization.^{22,23}

Somatosensory evoked potentials (SEP), and in particular the N20 response to peripheral nerve stimulation, reflect the electrical activity of different areas and tracts of the nervous system and correlate with cortical cerebral perfusion²⁴. SEP recording is a fast, non-invasive and reproducible technique that can be performed at the patient's bedside and provides real-time data on the functional status and perfusion of brain even when the patients are anesthetized or in an induced coma.²⁵

The PROMISE trial²⁶ investigated the presence and amplitude of the N20 response of SEP recorded before EVT in the ischemic hemisphere of 223 patients with anterior LVO undergoing EVT. N20 predicted functional independence at day 7 with a sensitivity of 93% (95%CI, 78-98%) and negative predictive value of 93% (80-98%), which increased to 100% after the EVT. Indeed, its predictive performance before treatment was superior to that of common factors used in clinical routine and added 0.07 points to the predictive model combining clinical variables, NIHSS score and ASPECTS (AUC of 0.84 versus AUC 0.77). Furthermore, N20+ had a higher predictive value of good outcome than advanced imaging variables recorded at baseline, such as the volume of infarct core, volume of hypoperfusion and collateral circulation. Taken together, these findings indicate that N20 might be a neurophysiological surrogate of salvageable brain and highlight the concept that ischemic brain tissue may remain viable from the perspective of cortical electrical function. Therefore, N20 SEP monitoring is a novel neurophysiological biomarker in acute stroke with the potential of having a major impact on the way stroke patients are managed throughout the world.

A major limitation of the PROMISE study was that N20 recording in the angio-room was interfered by artifacts precluding the blinded correct evaluation of the SEP in 25% of patients. These artifacts occurred in available commercial multifunction equipment not designed for the particular detection of N20 avoiding potential artifacts such as movements, electromagnetic interferences, noise, etc. Importantly, it also requires specialized-trained physicians to be used.

Time is Brain S.L. has developed a portable medical device, BraiN20®, which allows a fast (like ECG) non-invasive, real time and automated continuous monitoring of SEP that does not require special training.

PROMISE20 was the first in human study (NCT06149754) aiming to confirm the high reliability of the medical device BraiN20® automatic recording of N20. The trial was halted in July 2024 after a futility analysis of the first 17 patients studied due to usability difficulties of the device and electrodes. Tolerability was good and there were no side effects. Technical and usability improvements of BraiN20® device have been implemented and are now available for a second validation trial, the PROMISE20 RELOADED.

In this study we aim to investigate the reliability of the updated BraiN20® monitoring device to give an accurate diagnostic result of present N20 SEP (N20+), absent SEP N20 (N20-) or non-assessable result (noisy N20) according to the classification by two expert physicians blind to the BraiN20® reading results.

5.2. Trial design

Prospective, multi-center, single arm, interventional study with blinded evaluation of the principal secondary endpoint of a cohort of patients with AIS and anterior LVO undergoing EVT.

Patients will be included once AIS and anterior LVO are confirmed by CT/CTA or MR/MRA and when, at the physicians' criteria, EVT is indicated. A single SEP monitoring of 3-min with the BraiN20® medical device will be performed prior to the access to the angio-room or while the patients remain in the angio-room but before

the arterial puncture. A follow-up SEP monitoring will be performed after as soon as possible after EVT, within 30 minutes, and preferably in the angio-room.

Patients' cohort will be followed-up to 90 days after inclusion.

5.3. Clinical development plan

See complete clinical development plan in the attachment section below.

All the studies are being done according to *ISO 14155 Good Clinical Practice* and applicable regulations to the place where the study is carried out.

6. BENEFITS AND RISKS OF THE INVESTIGATIONAL DEVICE, CLINICAL PROCEDURE, AND CLINICAL INVESTIGATION

6.1. Anticipated clinical benefits

According to the current clinical evaluation from 25/Feb/2025 the clinical benefits are identified below.

The BrainN20® system has been designed and developed in accordance with internal requirements aiming to democratize and expand the recording and monitoring of SEPs, so that any patient with a neurological disease or condition can access the expected clinical benefits of this test:

MDR Clinical Benefit Category	BrainN20® Clinical Benefit	Expected Clinical Outcome (Patient-Level Impact)
1. Improved Neurological Monitoring & Diagnosis	Enables early detection of somatosensory dysfunction	Early identification of spinal cord injuries, stroke, multiple sclerosis (MS), and peripheral neuropathies
	Provides quantitative and objective neurophysiological data	Improves diagnostic accuracy compared to subjective clinical evaluations
2. Intraoperative Neuromonitoring (IONM) for Surgery	Provides real-time monitoring of somatosensory pathways during surgery	Reduces risk of intraoperative nerve damage
	Detects early signs of ischemia or mechanical nerve injury	Enables timely intervention by surgeons, improving surgical outcomes
3. Prognostic Value in Critical Care & ICU Monitoring	Assesses neurological function in comatose patients (e.g., after cardiac arrest or traumatic brain injury)	Predicts likelihood of recovery, aiding clinical decision-making
	Continuous monitoring in ICU settings allows detection of deterioration in neurological status	Enables timely medical interventions
4. Functional Monitoring in Neurodegenerative Diseases	Tracks progression of neurological diseases (e.g., MS, ALS, polyneuropathy)	Supports early diagnosis & intervention

MDR Clinical Benefit Category	BrainN20® Clinical Benefit	Expected Clinical Outcome (Patient-Level Impact)
	Helps monitor treatment effectiveness (e.g., rehabilitation, neuroprotective therapies)	Allows adjustments to therapy based on objective data
5. Non-Invasive & Safe Neurophysiological Monitoring	Uses surface electrodes with limited intact skin contact	Minimizes patient discomfort and risk of infections
	No ionizing radiation or contrast agents required	Safer alternative to CT/MRI for functional assessment
6. Optimization of Patient Management & Workflow	Portable design enables use in ambulance and emergency settings and facilitates its integration in hospital Critical Care areas	Expands neurophysiological monitoring capabilities outside hospitals
	Enables remote monitoring & telemedicine applications	Improves access to neurophysiological care in underserved areas

The BrainN20® system specific features that supports the feasibility of these clinical benefits are:

- Portable. The specialization of the BrainN20® system for SEP recording implies the reduction of its electronics. Furthermore, the display is integrated into the electronics, avoiding the need of the system to be connected to an external PC. The system allows a bedside SEP recording and monitoring.
- Fast and Ease to use.
The BrainN20® detection algorithm takes 33 seconds.
Intuitive and straightforward user interface aimed at a specific function.
No installation required.
- Single use consumables. One unique patient use. No cleaning between uses needed.
- Non-invasive, aiming to minimize risks to the patient and user and to expand the profile of potential healthcare personnel users.
 - Does not use ionizing radiation (compared to computed tomography imaging diagnostic techniques).
 - Does not use radiocontrast agents (compared to computed tomography with contrast agents: CTA and CT Perfusion and angio-Magnetic Resonance Imaging (angio-MRI)).
- Reliability >80%, according to the results of the non-clinical tests conducted as a part of the verification and validation process (see *Test n° 5.6 N20 reliability report*).

These intended clinical benefits related to the use of the device under evaluation has been supplemented with information found in literature searches or state of the art for the device category.

The available clinical data adequately show that the intended purpose of the device is achieved, thereby leading to a clinical benefit for patients, when the device is used as intended.

According to the PMCF Plan (see attachments section below 8), the anticipate clinical benefits are listed below.

This PMCFP will cover some minor changes and additional claims planned for BrainN20®.

Listed below minor design changes planned from V2.0 to V2.1:

- Elimination of the charging station and replace with internal charging.
- Stimulation electrodes: Textile support has been removed. Improving primary packaging to increase use by to 2 years.
- Briefcase for device, charger and connector is replaced by box.

The clinical evaluation covers device design features that pose special performance or safety concerns, the intended purpose and application of the device (e.g. target treatment group and disease, proposed warnings, contraindications, precautions, and method of application) and the specific claims made by the manufacturer about the clinical performance and clinical safety of the device.

A list of special characteristics is specified below:

Enable the measurement and automatic analysis of the N20 response of SEP during the AIS patient workflow:

- The BrainN20 medical device has good reliability to accurately identify the presence or absence of the N20 response of SEP in patients suffering from AIS due to LVO
- The BrainN20 algorithm predicts the functional outcome of AIS patients

6.2. Anticipated adverse-device effects

According to guideline *MDCG 2020-10/1* Safety reporting in clinical investigations of medical devices under the *Regulation (EU) 2017/745*, definitions of the relevant concepts are:

Adverse Event (AE):

Any untoward medical occurrence, unintended disease or injury or any untoward clinical signs, including an abnormal laboratory finding, in subjects, users or other persons, in the context of a clinical investigation, whether or not related to the investigational device.

Note:

- a. This definition includes events that are anticipated as well as unanticipated events*
- b. This definition includes events occurring in the context of a clinical investigation related to the investigational device, the comparator or the procedures involved.*

Serious Adverse Event (SAE):

Any adverse event that led to any of the following:

- a) death,*
- b) serious deterioration in the health of the subject, that resulted in any of the following:*
 - i. life-threatening illness or injury,*
 - ii. permanent impairment of a body structure or a body function,*

- iii. hospitalisation or prolongation of patient hospitalisation,
- iv. medical or surgical intervention to prevent life-threatening illness or injury or permanent impairment to a body structure or a body function,
- v. chronic disease,
- c) foetal distress, foetal death or a congenital physical or mental impairment or birth defect

ADE:

Adverse event related to the use of an investigational medical device

SADE:

A serious adverse device effect is an adverse device effect that has resulted in any of the consequence characteristics of a serious adverse event.

Device deficiency (DD):

Any inadequacy in the identity, quality, durability, reliability, safety or performance of an investigational device, including malfunction, use errors or inadequacy in information supplied by the manufacturer.

Based on the risk assessment performed on the BrainN20[®], device deficiency (DD) that could be related with the use of the device are:

1. Design / Component / Mechanical Deficiencies

- Device not charging or cannot power on.
- Display malfunction (e.g., leaking LCD, unreadable display).
- Electrode detachment during use.
- Non-uniform skin contact (acquisition or stimulation electrodes).
- Damaged or expired headband/stimulation electrodes used.
- Broken cable, connector, or housing.
- Use of incompatible or third-party components.
- Overheating of device.
- Charging during operation despite restriction.
- Fluid ingress into electronic parts.
- Unintended restart or shutdown.
- Electrode gel dried or improperly stored.

2. Use Errors (Human Factors)

- Incorrect electrode placement.
- Use error – action: user connects electrodes improperly.
- Use error – perception: user misreads LED or display.
- Use error – cognition: user misunderstands IFU warnings or sequence.

- Use of electrodes on inflamed/infected skin.
- Use by untrained personnel.
- Use on contraindicated population (e.g., pregnant, pacemaker patients).
- Simultaneous use with HF surgical equipment or defibrillator.
- Not monitoring skin during long stimulation periods.
- Improper disposal of single-use electrodes.

3. Software / Connectivity / Cybersecurity

- Software glitch (e.g., freeze during acquisition).
- Data loss or incomplete SEP acquisition.
- Delayed or failed Bluetooth/4G transmission.
- Insecure network connection (use on public Wi-Fi).
- Unauthorized software installed.
- Use with outdated firmware.
- Default passwords not changed (data confidentiality risk).

4. Electromagnetic Interference (EMI)

- Malfunction due to nearby RF equipment (e.g., electrosurgery).
- Interference with or from other medical devices.
- Non-compliance with EMC labeling or warnings.

These are failures, malfunctions, or non-conformities of the device, including use errors and cybersecurity issues.

Possible device defects (DD) that could be the cause adverse device effects (ADEs), could be, among others:

- Skin redness, itching, or irritation (mild).
- Localized discomfort from stimulation.
- Tingling or burning sensation during stimulation.
- Red marks or depressions after removing electrodes (transient).
- Mild allergic reaction to electrode adhesive or gel.
- Patient anxiety or discomfort due to stimulation sensation.
- Light dizziness or vertigo during or after testing.

These are unintended medical occurrences likely related to the device, not necessarily serious.

Serious adverse event that has a causal relationship with the investigational device, the comparator or the investigation procedure or where such causal relationship is reasonably possible are not expected (reportable SAEs). Listed below some possible SAEs:

- Blisters or ulcers: from excessive stimulation (burns).
- Cardiac stress: or arrhythmia in sensitive individuals.
- Muscle or nerve injury: due to prolonged or improper stimulation.

- Severe allergic reaction: or anaphylaxis to materials (rare).
- Skin infection: following breakdown of the skin barrier.
- Burns: due to simultaneous use with HF surgical unit.
- Data breach: leading to unauthorized access to personal medical data (GDPR impact).
- Incorrect clinical decision: due to failed SEP recording or artifact-contaminated data.

These are adverse events that result in serious outcomes (hospitalization, permanent damage, etc.).

6.3. Risks associated with the participation in the clinical investigation

Doctors will not make any specific decisions based on the results of the BrainN20® medical device, so routine stroke treatment will not be modified. Patients will be treated according to the European Stroke Organization (ESO) guidelines. Intravenous alteplase or tenecteplase will be administered if required before BrainN20® monitoring, which will not delay the thrombectomy workflow since monitoring may be also done in the angio-room while the arterial puncture is performed. There may be some restrictions in the local anesthetic protocols to avoid interferences with the N20 SEP.

6.4. Possible interaction with concomitant medical treatments considered under the risk analysis

The protocol does not imply the use of any additional medication.

6.5. Steps that will be taken to control or mitigate the risks

6.6. Rationale for benefit-risk ratio

On the one hand, the MD under test is considered not to add new risks with clinical impact to those already existing as part of the current medical procedure. Such risks are also not considered to become more probable with BrainN20® than without it. On the other hand, the BrainN20® provides an additional means to the current clinical practice.

Based on the above, there are no risks associated with the participation in the clinical investigation, neither possible interaction with concomitant medical or interventional treatments. Taking into account the anticipated clinical benefits derived from the use of BrainN20®, and the eventual device-related adverse effects, it can be concluded that expected benefits outweigh potential risks, and therefore, that the current benefit / risk profile for the device is favourable.

7. OBJECTIVES AND HYPOTHESES OF THE CLINICAL INVESTIGATION

7.1. Hypothesis

The BrainN20® medical device has good reliability to accurately identify the presence or absence of the N20 SEP in patients with AIS and LVO.

7.2. Objectives

The primary objective is to prove the reliability of the updated BrainN20® monitoring device to provide an accurate diagnostic result of present N20 SEP (N20+), absent N20 (N20-) or non-assessable result ("noisy" N20). Specifically, to study if BrainN20® automatic results of N20 SEP have an agreement with a kappa statistics >0.60 with the classification of the monitored SEP recording by two expert physicians blind to BrainN20® reading results and clinical outcome.

Secondary objectives are:

- The principal secondary objective is to determine the predictive performance of functional outcome at day 90 of an algorithm based on the presence and characteristics of the N20 response before EVT recorded through the BrainN20® device.
- To evaluate whether an interaction exists between the presence of N20 and recanalization in relation to the patient's outcome, as measured by the mRS.
- To study if BrainN20® provides at least a proportion of assessable patients (N20+ or N20-) equal or higher than 0.65 for a true proportion of 80%.
- To determine the rate of present N20 before EVT by clinical stroke severity (NIHSS score <17 or ≥17), time from symptom onset to first recording (<6h or ≥6h) and ASPECT score (<5, ≥5).
- To determine the rate of present N20 before EVT by dramatic early favorable response as determined by an NIHSS of 0-2 or NIHSS improvement ≥ 8 points at 24 (-/+ 6 hours) hours.
- To evaluate the predictive performance of the BrainN20® algorithm before EVT on functional outcome at day 7.
- To estimate the frequency of patients in whom present N20 SEP (N20+) recording prior to treatment has disappeared after the thrombectomy procedure, and the potential associated adverse events.
- To estimate the frequency of patients in whom absent N20 SEP (N20-) recording prior to treatment has appeared after the thrombectomy procedure.
- To evaluate the predictive performance of the BrainN20® algorithm after EVT on functional outcome at day 7 and 90
- To describe usability problems
- To register the number of consumables used
- To list frequency and causes of non-assessable recordings of N20.

Safety outcomes related to BrainN20® monitoring will be:

- Mortality at day 7
- Tolerability of the BrainN20® monitoring
- Device deficiency recordings
- Serious adverse events (related and not related to the monitoring procedure) within the study period.
- The time consumption in setting electrodes and monitoring before treatment.
- Door to puncture time in patients treated with EVT

8. DESIGN OF THE CLINICAL INVESTIGATION

8.1. Study setting

The PROMISE20 RELOADED study will be performed in 3 comprehensive stroke centers in Spain:

1. H.U. Vall d'Hebron, Barcelona, Spain. Reference EC.
2. H. U. Santa Creu i Sant Pau, Barcelona, Spain.
3. H. U. Bellvitge, Hospitalet (Barcelona), Spain.

8.2. Subjects

8.2.1. Inclusion Criteria

1. AIS with anterior LVO undergoing EVT treatment according to local protocols.
2. Age ≥ 18
3. No significant pre-stroke functional dependence ($mRS \leq 2$)
4. Patient treatable within 24 hours of symptom onset. Symptoms onset is defined as the time the patient was last seen well, and treatment point as the time of arterial puncture.
5. Baseline NIHSS score obtained prior to procedure must be equal or higher than 1 point
6. Occlusion (TICI 0-1) of the intracranial ICA (distal ICA or T occlusions), MCA-M1 segment or M2 suitable for EVT, as evidenced by CTA or MRA with or without concomitant cervical carotid occlusion or stenosis.
7. Any infarct core, without volume restrictions, measured by ASPECT score on non-contrast CT, or DWI-MRI prior to EVT
8. Informed consent obtained from patient or acceptable patient surrogate; or the deferred informed consent, to avoid the delay in the start of the mechanical thrombectomy.
9. Because this trial does not provide benefit and does not interact in treatment decision making during the trial, included patients may also participate in other clinical trials.

8.2.2. Exclusion Criteria

Clinical criteria

1. Patients with a well-documented history of neuromuscular diseases and other neurodegenerative disorders, prior stroke (TIA not excluded) or nervous system tumors that could interfere with SEP assessment.
2. Serious, advanced, or terminal illness with anticipated life expectancy of less than three months.
3. Patients with a pre-existing neurological or psychiatric disease that would confound the neurological or functional evaluations.
4. Women in the premenopausal period.

Neuroimaging criteria

5. CT or MR evidence of hemorrhage (the presence of microbleeds is allowed).
6. Subjects with occlusions in multiple vascular territories (e.g., bilateral anterior circulation, or anterior/posterior circulation)
7. Evidence of intracranial tumor (except small meningioma).

Brain20® medical device safety issues:

8. Subjects with a demand-type cardiac pacemaker, defibrillator, or other electrical implant or metal.
9. Patients with suspected or well-known cancerous skin lesion in the area where electrical stimulation will be applied.
10. Patients who have a localized disorder in the wrist and forearm where electrical stimulation is to be applied (i.e., fractures or dislocations).

8.2.3. Enrolment: Patients will be enrolled in a single arm and consecutively

The system is non-invasive and has no safety concerns. No interruptions neither for SAEs nor interim analysis are expected.

8.2.4. Duration of subject's participation

The participation of each subject enrolled in the investigation will encompass 90 days from enrolment.

8.2.5. Subject replacement

In case of subjects need to be replaced for any reason, enrolment will be continued in order to replace the missing subjects until completing the total of planned patients (n).

8.2.6. Lost to follow-up

Follow-up is foreseen to take place by a telephone call or on-site by a local assessor who will be blinded to acute phase results. Notwithstanding, in case lost to follow-up occurs, the patient will be contacted up to 5 times and until 20 days after the procedure. If no contact can be made, investigators can access the clinical file of the patient to find out possible reasons. Also, family can be contacted based on secondary telephone contact details as additional source of information.

8.3. Endpoints and outcome

The primary endpoint will be the BrainN20® monitoring result before EVT: present N20 SEP (N20+), absent SEP N20 (N20-) or non-assessable ("noisy" N20). Reliability of the BrainN20® Medical Device automatic result of N20 SEP will be analysed by concordance with the classification of the monitored SEP recording by two expert physicians blind to BrainN20® reading results and clinical outcome.

The principal secondary endpoint is the functional capacity at day 90 evaluated through a structured interview of the modified Rankin Scale performed by local assessors blinded to BrainN20® monitoring results and acute phase clinical outcome.

The cohort will be classified in three prognostic groups according to the mRS score at 90 days: (1) good functional recovery, mRS score 0-2; (2) mild functional recovery, mRS score 0-3; and (3) severe functional dependence or death, mRS score 4-6.

Other secondary endpoints are:

- The proportion of BrainN20® assessable patients N20+ or N20-.
- Rate of detection of N20+ before EVT by clinical stroke severity (NIHSS score <17 or ≥17), time from symptom onset to first recording (<6h or ≥6h) and ASPECT score (<5, ≥5).
- Rate of detection of N20+ before EVT by dramatic early favorable response as determined by an NIHSS of 0-2 or NIHSS improvement ≥ 8 points at 24 (-/+ 6 hours) hours.
- Frequency of patients in whom present N20 SEP (N20+) recording prior to treatment has disappeared after the thrombectomy procedure, and the potential associated adverse events.
- Frequency of patients in whom absent N20 SEP (N20-) recording prior to treatment has appeared after the thrombectomy procedure.
- Usability problems
- Number of consumables used
- Frequency and causes of non-evaluable recordings of N20.

Safety endpoints mes related to BrainN20® monitoring will be:

- Mortality at day 7.
- Tolerability of the BrainN20® monitoring.
- Device deficiency recordings

- Serious adverse events (related and not related to the monitoring procedure) within the study period.
- The time consumption in setting electrodes and monitoring before treatment.
- Door to puncture time in patients treated with EVT.

8.4. Blinding

SEP expert examiners will evaluate the accuracy of the N20 recording result before and after EVT. They will be blind to the results of the BrainN20® SEP monitoring, both prior and after EVT, neuroimaging variables and clinical outcome. N20 readings will be stored in a restricted field of eCRF to these examiners.

The principal secondary outcome endpoint, functional capacity evaluated by the mRS score at day 90, will be determined by face to face or telephone using a structured interview performed by a local certified rater blinded to the BrainN20® recordings.

8.5. Variables

8.5.1. At Baseline before EVT

- Age, sex, risk factors (diabetes mellitus, hypertension, dyslipidaemia, ischemic heart disease, atrial fibrillation, previous stroke, previous anticoagulation treatment), vital signs (serum glucose, blood pressure, body temperature), premorbid modified Rankin Scale score assessed by a certified neurologist at hospital admission, INR, in for patients under warfarin / acenocumarol.
- NIHSS score at admission and before the first BrainN20® recording.
- Site of occlusion before EVT treatment evaluated by CTA or MRA.
- Intravenous reperfusion therapy.
- ASPECT score on CT or DWI-MR.
- BrainN20® monitoring result before EVT: N20 SEP present (N20+); N20 SEP absent (N20-) or N20 SEP non-assessable ("noisy").

8.5.2. During the EVT procedure

- Final eTICI score.
- Procedural complications.
- NIHSS evaluation at the end of EVT (before BrainN20®) monitoring.
- Tolerability of patients and comfortability of interventionalists during BrainN20® recording
- After EVT BrainN20® monitoring result: N20 SEP present (N20+); N20 SEP absent (N20-) or N20 SEP non-assessable ("noisy").

8.5.3. During the ASU admission

- NIHSS at 24h ($\pm$ 6h) and at day 7 $\pm$ 1 or at discharge if earlier.
- CT scan at 24-36h (local evaluation):
 - ASPECTS score
 - Hemorrhagic transformation (IH1, IH2, PH1, PH2, rPH)

8.5.4. After discharge

- Ischemic stroke etiology:
 - Large artery atherosclerosis (extracranial or intracranial).
 - Atrial Fibrillation (AF) or other cardioembolic diseases.
 - Unknown.
 - Other (e.i. arterial dissection).
- Modified Rankin Scale score at 7 $\pm$ 1 days (per blinded local assessor) and 90 $\pm$ 7 days (evaluated by telephone call or on-site by a local assessor who will be blinded to acute phase results).
- Device deficiency recordings (according to *section 16* below).
- Serious adverse events (related and not related to the monitoring procedure) within the study period.

8.5.5. Times

- Time of stroke onset and time of hospital admission.
- Time needed for setting up the electrodes, and to obtain the first BraiN20® recording.
- Time of the first BraiN20® recording.
- Time of groin/radial puncture.
- Time of complete reperfusion (eTICI 2b or 3).
- Time of the last BraiN20® recording after EVT (in the angio-room).

8.5.6. Participant timeline

Time point	Baseline		EVT		Follow-up at study hospital	
	t0	t1 BrainN20® first recording	t2 BrainN20® recording after recanalization/ at the end of procedure	t3 (24h from first recording date and time +/- 6h)	t4 (7 days from first recording date and time (+/- 24h) or discharge, whatever occurs first)	t5 (90 days from first recording date and time +/- 7d)
Admission details	X					
Eligibility criteria	X****					
Plane CT, or DWI-MR (ASPECTS score at baseline)	X			X		
CTA or MRA	X					
Informed consent	X*	X*	X*	X*	X*	X*
NIHSS score	X	X	X	X	X	
Time of needed to set up electrodes and obtain the first BrainN20® signal		X				
Time of the first BrainN20® monitoring		X				
BrainN20® monitoring: N20+, N20- or non-assessable		X	X			
Time of groin/radial puncture		X				
Time of reperfusion (eTICI 2b-3)			X			
Final eTICI score			X			
Adverse events		X	X	X	X	X
mRS					X***	X**

X*- To be obtained at Baseline or during hospitalization (if deferred consent approved by the EC).

X**- Modified Rankin Scale score at 90 days by blinded local investigator, telephone or onsite.

X***- Modified Rankin Scale score by blinded local investigator.

X**** In case of women of childbearing potential (WOCBP) with no highly effective methods for birth control, a pregnancy test should be performed.

8.6. Investigational device

- SEP monitoring will be carried out using the BrainN20® medical device and appropriate electrodes. Scalp electrodes will be embedded in a headband and wrist electrodes stucked-on both wrists of the patients. The device provides an automatic reading of N20 and outcome prediction and do not require a specific training. Therefore, examiners will be physicians or nurses in charge of the patient. A user's manual will be provided to the investigators and specific training sessions will be organized.

2. SEP of both median nerves will be recorded, transferred and stored to the EAC for their evaluation. BrainN20® Medical Device provides an automatic reading of the presence and amplitude of an N20 response ipsilateral to the cerebral hemisphere affected by the stroke and also of the contralateral hemisphere as control. The BrainN20® detection algorithm includes a set of conditions that automatically define the presence or absence of N20, as well as its amplitude and latency. The device measures one N20 signal per 33 seconds and gives an average result in 3 minutes. Three result messages may appear on the screen: “N20 present”, “N20 absent” or “N20 not-assessable”.
3. Registries will be recorded in a SD memory card that is included inside the device. Data will be downloaded from the SD card periodically and transmitted to the expert physicians for reliability evaluation of recordings, blind to the BrainN20® results.
4. The reliability of N20 recordings is classified in 3 categories: “N20 present”, “N20 absent” or “N20 not-assessable”. Appropriate BrainN20® recording by investigators will be checked in three pilot patients per site before starting the trial. These patients will be added to the sample size and considered only for safety, but not for the rest of analysis.

8.7. Procedures

At Admission at the Emergency Unit, acute stroke patients with suspected LVO will be immediately attended by neurologists and transferred to the CT/MR room for neuroimaging evaluation. If an ischemic stroke is confirmed and the patient is eligible for IV-tPA or TNK, IV thrombolysis will be administered without delay. CTA or MRA will be performed to confirm anterior LVO.

Serum glucose, body temperature and blood pressure will be controlled following the ESO guidelines.

ASPECTS score will be measured by local investigators on CT or DWI-MR but it will not preclude EVT if this is indicated according to local protocols. Perfusion CT or MR is advisable but not mandatory.

The patient will be transferred to the angio-suit as soon as possible for catheterization and EVT. Prior to puncture, BrainN20® electrodes will be set up, and NIHSS score and the result of the first BrainN20® reading will be recorded in the CRF. Informed consent will be obtained from the patients or family members. In addition, oral and deferred Informed consent has been accepted by the referral Ethical Committee (according to the section below 15).

Baseline SEP recording will be performed in the emergency room or in the angiography room before the puncture. During the EVT procedure, the system may be removed if it is required by the neurointerventionists. However, the head band should not be removed to ensure that the gel and electrodes remain in the same position that at baseline monitoring avoiding artifacts.

During endovascular procedure, EVT with or without prior intravenous thrombolysis will be performed with stent retrievers, thrombus aspiration systems or combined technique. According to the evidence currently accumulated, sedoanalgesia with spontaneous ventilation is preferred over orotracheal intubation and

general anesthesia. If general anesthesia is performed, propofol (Target-controlled infusion (TCI), 4 mcg/ml) and remifentanyl (TCI, 4 ng/ml or 0.03-0.015 ng/kg/x') will be used for induction, and maintenance will be carried out with a continuous infusion of propofol (TCI, 2-3 mg/ml) and remifentanyl (TCI, 3-7 ng/ml). Muscle relaxants will not be used throughout the procedure, except for the general anesthesia (0.5 mg/kg rocuronium). Halogenated anaesthesia as well as benzodiazepines should be avoided before and during the endovascular procedure (Mirallave et al., 2021, DOI: [10.1007/s10877-020-00621-9](https://doi.org/10.1007/s10877-020-00621-9))

Serum glucose will be determined at baseline and blood pressure, heart rate, body temperature and oxygen saturation will be continuously monitored during the EVT procedure. Subsequently, patients will be admitted to the acute stroke unit or intensive care unit and treated according to the clinical practice guidelines of the ESO.

After endovascular thrombectomy, the time and the result of the BrainN20® reading at the end of the procedure will be recorded in the CRF. The end of the EVT procedure will be at the discretion of the interventionalists. The final eTICI score and NIHSS score will be recorded.

Because of neurological damage can occur during the procedure and after reperfusion, SEP monitoring will be repeated after the EVT as soon as possible within the next 30-minutes, preferably in the angio-room. Thereafter, electrodes will be removed, and the patients will be transferred to the ASU of ICU as appropriate.

8.8. Data monitoring

The Steering Committee has assigned a CRO to this study whose duties are to aid the P.I. and the Steering Committee members in the maintenance of complete, legible, well organized, and easily retrievable data. Personnel from CRO will ensure that the study complies with relevant Good Clinical Practices (GCPs). Periodic monitoring visits will be made throughout the investigation to ensure that the investigator's obligations are being fulfilled. The PROMISE20 RELOADED database will be kept confidentially at the CRO for centralized monitoring of the electronic records.

Any data discrepancies may be queried during ongoing review of data by the Study Coordinating Center or may be identified and queried during routine monitoring visits. Monitoring visits will be performed to verify data accuracy and ensure queries are resolved.

The PROMISE20 reloaded study protocol will be conducted according to a Monitoring Plan aiming to:

- Ensure that the monitoring procedures are constant and global during the study.
- Provide a consistent approach to the conduct of the research study among investigational sites.
- Assure completion of the project in a timely manner to the highest standards of quality.
- Provide consistent study guidance to the study CRAs.

9. STATISTICAL DESIGN AND ANALYSIS

9.1. Sample size

The proportion of patients with accomplished baseline BrainN20® recording is estimated in 80%. Following previous results, we estimate a real rate of present N20 of 49%, absent N20 of 26% and non-assessable N20 SEP of 25% in acute stroke patients with LVO before EVT. Accordingly, assuming a real kappa (K) value equal to 0.75 and in order to prove an agreement of $K > 0.60$ using the lower limit of the 95%CI, 72 patients will be needed. No interim analysis will be done.

9.2. Statistical considerations

- The following section describes the statistical methods for the clinical investigation. Additional details on statistical analyses, including justification of clinical investigation design, sensitivity analyses, probability analyses, subgroup analyses and analyses of descriptive endpoints, may be maintained in a separate Statistical Analysis Plan (SAP).

9.3. Analysis Populations

- Screening patients: informed consent signed.
- Intention to test population: patients fulfilling the inclusion and exclusion criteria with accomplished baseline BrainN20® monitoring.
- Per protocol population: Intention to test population with assessable N20 SEP monitoring (exclude non-assessable patients).

Reasons of exclusion of every population will be detailed.

9.4. General statistical approach

- Continuous data will be summarized using:
- The number of non-missing / missing data, mean, standard deviation (SD), median, and minimum, Q1, Q3, and maximum.
- 95 % confidence interval of the mean will be provided when relevant.
- Categorical and ordinal data will be summarized using:
- The number of non-missing/missing data, counts and percentage.
- Missing data or unknown responses will not be included in the percentages.
- 95 % confidence interval will be provided when relevant.

9.5. Primary Analyses

The primary objective will be analysed in the Intention to Test Population. Response categories recordings and from experts classification (N20 present, N20 absent, N20 non-assessable) will be summarize by

frequencies and percentages. Agreement will be quantified by the Kappa (**K**) statistic of each examiner with the BrainN20® results (3 categories: N20+, N20- and noisy) and 95%CI will be calculated. In addition, interrater agreement will be estimated.

9.6. Secondary Analyses

The principal secondary endpoint is to obtain a prognostic performance of an algorithm based on the presence and characteristics of the N20 response of SEP recorded through the BrainN20® on functional outcome at day 90 in patients with AIS and anterior LVO undergoing EVT. This analysis will be performed in the Per Protocol population. The proportion of poor functional recovery will be calculated for both groups (N20+ and N20-). The predictive performance of present N20 SEP before EVT as a biomarker of good functional recovery, mild functional recovery and severe stroke dependency or death will be analysed by logistic regression and quantified as odds ratio with 95 % confidence interval compared to patients with the presence of N20 SEP. The predictive effect of N20 on functional recovery will be further measured by the metrics of sensitivity, specificity, positive predictive value, and negative predictive value with 95 % CI.

9.7. Subgroups analysis

- Presence/Absence of N20 might be used as strata in specific analysis. Other parameters will be defined on statistical analysis plan.

9.8. Multiple comparison / Multiplicity

- No multiple comparison is applicable since the differences will be processed as binary.
- The secondary endpoints as outlined in 8.3 *Endpoints and outcome* will be considered as supportive only. Therefore, no alpha-adjustment for multiple endpoint testing will be applied.

9.9. Pooling Strategy

- Separate analysis by stroke centers might be performed.

9.10. Missing data

- Missing imputation strategy will be detailed on statistical analysis plan. Example: parameters missings as NIHSS score and mRS score after patients die could be imputed to last observation carried forward (LOCF).

9.11. Variables transformation

Some variables whose distribution is skewed might be transformed. Example: amplitude of N20 response log-transformed. Other specific, will be detailed on statistical analysis plan.

 timeisbrain	CLINICAL INVESTIGATION PLAN PROMISE20 RELOADED	CODE: PROMISE20 RELOADED CIP REV: 1 DATE: 11/04/2025 PAGE: 32 of 44
--	---	---

9.12. Interim analysis

No interim analysis is planned.

9.13. Independent committees / labs

N20 SEP Core Lab Committee

Two expert independent physicians blind to the acute stroke variables and outcome will review the BrainN20® monitoring results before and after the EVT procedure. Results will be classified as N20 Present, N20 absent, N20 non-assessable. Local investigators will record the N20 amplitude and technical characteristics of monitoring quality such as impedance, stimulus intensity and misusing problems. The predictive proportion of functional recovery at day 7 and 90 will also be registered in an independent database.

10. DATA MANAGEMENT

10.1. Data collection methods

The data will be collected on an electronic case report form (CRF) designed specifically for this purpose (for more details refer to the attachment section below 5).

The completion of the eCRF will be made by authorized site personnel. Each authorized member will be provided with a unique, non-transferable username and password to allow access to the eCRF.

The investigator must ensure the accuracy, completeness and timeliness of data reported in the eCRF. Any change or correction in the eCRF will be tracked through an audit trail procedure.

The eCRF will include electronic validation checks for data verification. Validation procedures (edit checks) will run on new or modified patient data.

The program will test all data for completeness, consistency and plausibility, and produce queries for erroneous, incomplete and missing data.

An eCRF will be completed for each patient who is assigned a study number and after completion of the informed consent process. All trial data will be entered on the eCRF.

Study personnel at each site will enter data from source documents into the eCRF. Patients will not be identified by name in the study database or on any study documents to be collected by the Sponsor (or designee) but will be identified by a patient number.

Any necessary queries are sent to the investigator by the CRA via eCRF or email. Resolved queries will be returned to the CRA via eCRF (or email). If answers are incomplete or discrepant with other data, re-queries may be necessary.

The Data Management group will be responsible for data processing, in accordance with procedural documentation. Database lock will occur once quality assurance procedures have been completed.

11. AMENDMENTS TO THE CIP

This clinical investigation plan shall be amended using document control best practices, in accordance with quality management system (QMS) procedures established at Time is Brain, S.L. Any change shall be identified and briefly described in *section Revision History*, and the corresponding approved new version of the CIP shall be signed by principal investigators. Relevant amendments will be submitted and approved by the Ethics Committee and the Spanish Competent Authority (Agencia Española del Medicamento y Productos Sanitarios – AEMPS) prior their implementation.

12. DEVIATIONS FROM CLINICAL INVESTIGATION PLAN

Neither the Principal Investigator nor any other personnel involved in the conduct, monitoring and analysis of the clinical investigation shall perform any action that deviates from this clinical investigation plan.

All protocol deviations may be identified and collected by any project team member including CRO, sponsor, Investigational site personnel and/or vendor teams and will be documented from project start to project closure. A divergence from the Investigational Plan or study specific instructions that has the potential to significantly affect subject rights/safety or data integrity of trial results (major protocol deviation), will be duly notified to the corresponding regulatory statements CEIm and AEMPS. Minor protocol deviations will be notified to the ethics committee and Regulatory Agency in the safety follow-up report, as applicable.

13. DEVICE ACCOUNTABILITY

Investigational devices will be provided by the sponsor to the site. Accompanying packing list shall be signed by the PI or delegation as reception and retained in the ISF.

Sufficient safety stock will be available to ensure continuity of supply.

Investigation devices shall be stored according to the provided instruction of use.

Only the PI and clinical investigation personnel will be allowed to have access to the device.

Any device deficiency shall be registered according the section below 16. Non-conforming products shall be segregated as such and returned to the sponsor for investigation.

Investigation devices traceability is assured through the packing list and the e-CRF. Stock reconciliation shall be reviewed during monitoring visits.

14. STATEMENTS OF COMPLIANCE

14.1. Initiation of the study

The clinical investigation will not be initiated until the due authorisation from the accredited Ethics Committee (CEIm) and the national competent authority (AEMPS) shall been obtained.

Single and binding favorable opinion of the CEIm shall be obtained. and all the participating centers will be informed.

By signing sponsor's protocol signature page, the principal investigator has documented his/her intention, commitment, and obligation to conduct the trial in compliance with GCPs, the applicable regulations and guidelines, and the Investigational Plan.

14.2. Ethical principles

The study will be conducted in compliance with the *Declaration of Helsinki (Fortaleza, Brasil, octubre 2013)* and in compliance with this CIP as well as with the *Regulation EU/2017/745 of 5 April 2017 on medical devices, as amended (MDR, Articles 62 to 80 and Annex XV)*, the MDR transposition into Spanish legislation in *Real Decreto 192/2023 of March 21 which regulates medical devices*, the *Royal Decree 1090/2015, 4 December, regulating the clinical trials with drugs, Ethics Committees and the Spanish Register of Clinical Studies*, and with the principles of good clinical practice as described in *Annex A of standard ISO 14155:2020 – Clinical investigation of medical devices for human subjects — Good clinical practice*.

Willingness to participate in the study will be ethically sought and signature of the informed consent will be obtained for each patient prior to enrolling in the study (refer to section **INFORMED CONSENT PROCESS**¹⁵).

14.3. Study financing

The study is financed by Time is Brain, S.L. and the European Innovation Council, Horizon-EIC-2022-Accelerator-01, project Promise 190126352.

14.4. Contracts

A contract between Time is Brain, S.L. and hospital is in place.

A contract between Time is Brain, S.L. and financing entity is in place.

14.5. Insurance

A legal insurance coverage for all injuries that may be the result of the clinical study measures performed on patients will be contracted.

14.6. Data protection

The processing, communication and transfer of personal data of all participants will comply with the provisions in *Regulation EU 2016/679 of the European Parliament and of the Council of April 27 2016 regarding the protection of natural persons insofar as to the processing of personal data and the free circulation of data*, the *Organic Law 3/2018, of December 5, on the Protection of Personal Data and guarantee of digital rights (LOPDPGDD)* and with the applicable national regulations in each of the participating countries.

The legal basis that justifies the processing of any personal data is the consent given in the Informed Consent Form, in accordance with the provisions of article 9 of *EU Regulation 2016/679*.

Data are pseudo-anonymized: each patient will be assigned a unique patient study number which does not allow identification of said patient. The link between the patient personal identification and the assigned patient number will be documented in a Patient Identification Log, which shall be securely filed in the investigator's study files (ISF) and kept at the hospital at all times.

The data collected for these studies will be identified only by a code, so that no information will be included that allows the participants to be identified. Only the principal investigator and her collaborators with the right of access to the data source (clinical history) may relate the data collected in the study with the patient's clinical history.

The identity of the participants will not be accessible to any other person except for a medical emergency or legal requirement.

Health authorities, the Ethics Committee and personnel authorized by the study sponsor may have access to identify personal data when necessary to verify study data and procedures, but always maintaining confidentiality in accordance with current legislation.

Only the encrypted data will be transferred to third parties and other countries, which in no case will contain information that can directly identify the participant (such as name and surname, initials, address, social security number, etc.). If this transfer occurs, it would be for the same purpose of the study described and guaranteeing confidentiality.

If a transfer of encrypted data is performed outside the EU, whether in entities related to the hospital centre where the patient participates to service providers or to researchers who collaborate with the sponsor, the data of the participants will be protected by safeguards such as contracts or other mechanisms established by data protection authorities.

As Sponsor of the study, Time is Brain, S.L., undertakes to process the data in accordance with *EU Regulation 2016/679* and, therefore, to keep a record of the processing activities that are carried out and to perform a risk assessment of such processing to know what measures will have to be applied.

In addition to the rights that the previous legislation already contemplates (access, modification, opposition and cancellation of data, deletion in the new Regulation), now study participants can also limit the processing of data collected for the study that are incorrect, can request a copy or that data be transferred to a third party (portability). To exercise these rights, participants must contact the principal investigator of the study, or the Data Protection Delegate of the hospitals involves. They also have the right to contact the Data Protection Agency if they are not satisfied.

Data cannot be deleted even if a patient withdraws from the study, to ensure the validity of the research and to comply with legal duties.

The Principal Investigator and the Sponsor are obliged to keep the data collected for the study for at least 10 years after its completion. Subsequently, personal information will only be kept by the health care centre and

by the Sponsor for other scientific research purposes if the patient has given its consent to do so, and if permitted by applicable law and ethical requirements.

15. INFORMED CONSENT PROCESS

The process of obtaining the informed consent form signed by each patient willing to participate in the study will be conducted in accordance with good clinical practices, following the general procedure described in standard *EN ISO 14155*.

Patients will be informed of all aspects of the investigation that are relevant for their decision to participate in the study.

- a) all aspects of the clinical investigation that are relevant to the subject's decision to participate throughout the clinical investigation will be explained;
- b) any coercion or undue improper influence on, or inducement of, the subject to participate will be avoided;
- c) subject's legal rights will not be waived;
- d) non-technical language that is understandable to the subject will be used;
- e) ample time for the subject to read and understand the informed consent form and to consider participation in the clinical investigation will be provided;
- f) it will be ensured that personally dated signatures of the subject and the principal investigator or an authorized designee responsible for conducting the informed consent process are obtained;
- g) the subject will be provided with a copy of the signed and dated informed consent form and any other written information;
- h) it will be ensured that documentation of the process is kept in the subject's source documents and that the investigation site's signed informed consents is kept with the essential documents;
- i) show how informed consent is obtained and recorded in special circumstances where the subject is unable to provide it him- or herself;
- j) it will be ensured that important new information is provided to new and existing subjects throughout the clinical investigation, which may relate to the subject's willingness to continue participation in the clinical investigation.

Deferred Informed Consent (DIC) will be allowed, if it is accepted by the IRB (no safety issues were observed in PROMISE20 study), in order to avoid any delay in the initiation of EVT. The DIC will be signed by the patient or next of kin at any time after the hospital admission occurred.

Alternatively, Informed consent will be obtained (written or oral consent) at study hospital from subject or acceptable subject surrogate (i.e. next of kin, or legal representative) prior the use of the medical device. If oral informed consent is given (by the patient or surrogate), informed written consent procedure will be reattempted during admission or follow up visits.

In patients without an acceptable patient surrogate at admission, the delegated person may explain the study to the patient surrogate by phone, obtain oral consent and document the discussion in the medical records. Upon arrival of a patient surrogate to the hospital, a signature on the informed consent form will be obtained confirming time and date of prior, oral consent given.

If they agree to enter the study, the subjects will receive instructions from the investigators which must be followed strictly. They also shall be instructed how to promptly contact the investigators (details for outpatient periods shall also be provided) in case any incident occurs during the study period

In case the patient is able to understand but cannot sign the ICF, a witness can complete, sign and date the specific section of the ICF.

In those cases where the original consent was given by an acceptable patient surrogate (i.e. next of kin, or legal representative), the patient, his/her clinical status allowing, will be requested to confirm continued consent in writing and in the case that, at the final follow-up visit, the clinical status of the patient still does not allow for the patient to confirm continued consent, the investigator will appropriately document this condition.

Once the ICF is signed, the Investigator will provide patients and, if applicable, patient surrogate with a copy of their signed ICF.

The Investigator will maintain a copy of the EC-approved ICF along with a copy of each patient's signed ICF in a designated study file. If the patient or surrogate has signed the ICF but is found not eligible for inclusion in the study, the patient will be considered Screening Failure. The informed consent procedure will be documented in the patient's Medical History.

16. ADVERSE EVENTS, ADVERSE DEVICE EFFECTS, AND DEVICE DEFICIENCIES

See section below 6 *OP.8.2-03 Vigilance system*. for vigilance process description.

The sponsor shall fully record all of the following:

- any adverse event of a type identified in the clinical investigation plan as being critical to the evaluation of the results of that clinical investigation.
- any serious adverse event.
- any device deficiency that might have led to a serious adverse event if appropriate action had not been taken, intervention had not occurred, or circumstances had been less fortunate.
- any new findings in relation to any event referred above.

Upon request by any Member State in which the clinical investigation is being conducted, the sponsor shall provide all information above.

The sponsor shall report, without delay to both the CEIm and the AEMPS of:

- any serious adverse event that has a causal relationship with the investigational device, the comparator or the investigation procedure or where such causal relationship is reasonably possible.
- any device deficiency that might have led to a serious adverse event if appropriate action had not been taken, intervention had not occurred, or circumstances had been less fortunate.
- any new findings in relation to any event referred above.

The period for reporting shall take account of the severity of the event. Where necessary to ensure timely reporting, the sponsor may submit an initial report that is incomplete followed up by a complete report.

- Imminent risk of death, serious injury, or serious illness and that requires prompt remedial action for other patients/subjects, users or other persons: Immediately, but not later than 2 calendar days after awareness by sponsor.
- Any other reportable events: Immediately, but not later than 7 calendar days following the date of awareness by the sponsor.

Tabular form “MD_MDCG_2020-10-2_guidance_safety_report_form_EN” shall be used.

The sponsor shall implement and maintain a system to ensure that the reporting of the reportable events will be provided by the investigator to the sponsor immediately, but not later than 3 calendar days after investigation site study personnel’s awareness of the event. e-CRF also has a AEs and DDs reporting module.

Investigator will intake immediately (24 hours of knowledge) all the detected DDs and AEs attaching related source documentation in the e-CRF. Monitor and Sponsor will receive a notification for immediate information revision and analysis.

See section above 6.2 for anticipated DDs, ADEs and SADEs.

All (S)AEs of special interest: All AEs of special interest will be described in a specific study guideline. The list of AEs of special interest will include the following events:

- Recognised Device or Procedural Complication.
- Arterial perforation during the procedure.
- Arterial dissection during the procedure.
- Thrombus embolization in a different territory during the procedure.
- Vasospasm during the procedure.
- Groin hematoma.
- Neurological worsening during the procedure (increase ≥ 4 points from the admission NIHSS score).
- Neurological worsening persisting for more than 24 hours during hospitalization (increase ≥ 4 points): Please report as neurological deterioration the clinical worsening not attributable to Haemorrhagic Transformation or thrombus embolization
- SICH (PH1/PH2/rPH associated to a NIHSS worsening of ≥ 4 points) during hospitalization
- Asymptomatic hemorrhagic transformation

- Recurrent stroke
- Malignant stroke
- Decompressed craniectomy
- Aspiration pneumonia
- Pulmonary embolism
- Seizures
- Atrial Fibrillation
- Congestive heart failure

17. VULNERABLE POPULATION

Not applicable, as the study does not involve recruitment of vulnerable population.

18. SUSPENSION OR PREMATURE TERMINATION OF THE CLINICAL INVESTIGATION

Subjects will be withdrawn from the study if one of the following occurs:

- The subject voluntarily discontinues his or her participation in the study.
- The investigator may prematurely discontinue the participation of any patient in the study if the investigator feels that the patient can no longer fully comply with the requirements of the study or if any of the study procedures are deemed potentially harmful to the patient.
- The subject was found to be not-eligible following signature of the Informed Consent Form

The reason for early discontinuation will be documented in the source documents and case report forms. Data collected up to the point of withdrawal or premature discontinuation will be included in the data analysis, unless either the study procedure was not performed with the investigational device or specifically requested (by the subject) not to.

The study documentation will be kept for at least 10 years after the clinical investigation with the device in question has ended, or, in the event that the device is subsequently placed on the market, at least 10 years after the last device has been placed on the market.

The end of the study will be notified to the AEMPS within 15 calendar days.

Temporary discontinuation or early termination will be notified to the AEMPS and a justification will be provided within 15 calendar days or 24 hours (in case of security reasons).

19. PUBLICATION POLICY

Protocol report

The protocol will be summarized and published in the clinicaltrials.org website.

Primary (Main Results) Paper

The PROMISE20 RELOADED main results will be published following trial completion and data analysis. The manuscript will include data on the primary efficacy and safety endpoints. The first draft of the manuscript will be prepared by the two co-PI (Antoni Dávalos & Alicia Martinez-Piñero) and will be revised and approved by the SC members. Authorship of the primary results paper will be the SC and independent committee members. Study Principal Investigators will be lead (first) author (Alicia Martinez-Piñero) and senior (corresponding) author (Antoni Dávalos). All enrolling sites, together with each site's PI (and sub-investigators where journal policy allows), will be listed in the paper's appendix in order of highest enrollment.

Additional Policies

- All PROMISE20 RELOADED data is proprietary of the sponsor and may not be shared in whole or in part without the permission of the Time is Brain Company.
- The PROMISE20 RELOADED main study results will be presented to the Investigators prior to presentation. These results are confidential, and all investigators will be asked to sign a confidentiality agreement at that time.
- Time is Brain will have no role in the predefined analyses, data interpretation, manuscript writing or decision to submit results, that will be decided by the SC members. However, the main paper will be submitted to the Company at least 15 days before for approval.
- All publications and presentations must acknowledge Time is Brain Inc.
- No presentation of study results in whole or in part is allowed prior to publication of the main results manuscript outside of the methods manuscripts described above.
- All press releases and media materials must be reviewed and approved by Time is Brain.
- The senior author and principal investigator are responsible for:
 - i. Integrity of data analyses
 - ii. Maintaining timeline for publication
 - iii. Assignment of tasks and writing assignments
 - iv. Compliance with publications policies and procedures as described in this document.
 - v. Correspondence with co-authors

Communication to the competent authorities:

The final report with the investigation results will be sent to the AEMPS within one year from the end of the investigation, or three months after its early completion or temporary discontinuation.

An annual report on the progress of the clinical investigation will be submitted to the AEMPS when the duration of the investigation exceeds one year.

20. TRAINING OF INVESTIGATOR AND SITE PERSONNEL

The training of the Investigator, and appropriate clinical site personnel will be the responsibility of the Study Coordinating Group and may be conducted during local investigator meetings, a site initiation visit, or other appropriate training sessions. Training will include, but not be limited to, the study protocol, eCRF completion, neurological scale evaluation, and site personnel responsibilities. All Investigators and site personnel who are trained must have their training documented. Prior to the initiation of the study and subject enrollment, the Study Coordinating Group or designee will visit each site where the trial is conducted. The Sponsor or designee will ensure that the site personnel are informed about and understand the clinical study requirements.

Specific training on BrainN20® electrodes set up and monitoring, will be offered to nurses and neurologists involved in the acute stroke management. It is expected that this monitoring will be as simple as an EKG. Each center will be provided with the portable equipment and the electrodes for the studied patients. The first three patients studied in each center will be considered *pilot* and will serve to check the quality of the monitoring by the CRO. These patients will not be included in the analysis.

21. BIBLIOGRAPHY

1. Sacks D, Baxter B, Campbell BCV, Carpenter JS, Cognard C, Dippel D, Eesa M, Fischer U, Hausegger K, Hirsch JA, et al. Multisociety Consensus Quality Improvement Revised Consensus Statement for Endovascular Therapy of Acute Ischemic Stroke. *American Journal of Neuroradiology*. 2018, 39 :E61-E76.
2. Goyal M, Menon BK, van Zwam WH, Dippel DW, Mitchell PJ, Demchuk AM, Dávalos A, Majoie CB, van der Lugt A, de Miquel MA, et al; HERMES collaborators. Endovascular thrombectomy after large-vessel ischaemic stroke: a meta-analysis of individual patient data from five randomised trials. *Lancet*. 2016;387:1723-31.
3. Venema E, Roozenbeek B, Mulder MJHL, Brown S, Majoie CBLM, Steyerberg EW, Demchuk AM, Muir KW, Dávalos A, Mitchell PJ et al; HERMES collaborators and MR CLEAN Registry Investigators*. Prediction of Outcome and Endovascular Treatment Benefit: Validation and Update of the MR PREDICTS Decision Tool. *Stroke*. 2021;52:2764-2772.
4. Bourcier R, Goyal M, Muir KW, Desal H, Dippel DWJ, Majoie CBLM, van Zwam WH, Jovin TG, Mitchell PJ, Demchuk AM, et al; HERMES Trialists Collaboration. Risk factors of unexplained early neurological deterioration after treatment for ischemic stroke due to large vessel occlusion: a post hoc analysis of the HERMES study. *J Neurointerv Surg*. 2022; Feb 15: neurintsurg-2021-018214.

5. van de Graaf RA, Samuels N, Chalos V, Lycklama A, Nijeholt GJ, van Beusekom H, Yoo AJ, van Zwam WH, Majoie CBLM, Roos YBWEM, et al; MR CLEAN Registry investigators. Predictors of poor outcome despite successful endovascular treatment for ischemic stroke: results from the MR CLEAN Registry. J Neurointerv Surg. 2022;14:660-665.
6. Li H, Huang J, Ye S, Chen H, Yuan L, Liao G, Du W, Li C, Fang L, Liu S, et al; DIRECT-MT investigators. Predictors of mortality in acute ischemic stroke treated with endovascular thrombectomy despite successful reperfusion: subgroup analysis of a multicentre randomised clinical trial. BMJ Open. 2022;12:e053765.
7. Lansberg MG, Mlynash M, Hamilton S, Yeatts SD, Christensen S, Kemp S, Lavori PW, Ortega-Gutierrez S, Broderick J, Heit J, et al; DEFUSE 3 Investigators. Association of Thrombectomy With Stroke Outcomes Among Patient Subgroups: Secondary Analyses of the DEFUSE 3 Randomized Clinical Trial. JAMA Neurol. 2019;76:447-453..
8. Bhuva P, Yoo AJ, Jadhav AP, Jovin TG, Haussen DC, Bonafe A, Budzik RJ, Yavagal DR, Hanel RA, Hassan AE, et al; DAWN Trial Investigators. Noncontrast Computed Tomography Alberta Stroke Program Early CT Score May Modify Intra-Arterial Treatment Effect in DAWN. Stroke. 2019;50:2404-2412.
9. Dutra BG, Tolhuisen ML, Alves HCBR, Treurniet KM, Kappelhof M, Yoo AJ, Jansen IGH, Dippel DWJ, van Zwam WH, van Oostenbrugge RJ, et al; MR CLEAN Registry Investigators. Thrombus Imaging Characteristics and Outcomes in Acute Ischemic Stroke Patients Undergoing Endovascular Treatment. Stroke. 2019;50:2057-2064.
10. Liebeskind DS, Saber H, Xiang B, Jadhav AP, Jovin TG, Haussen DC, Budzik RF, Bonafe A, Bhuva P, Yavagal DR, et al; DAWN Investigators. Collateral Circulation in Thrombectomy for Stroke After 6 to 24 Hours in the DAWN Trial. Stroke. 2022;53:742-748..
11. Campbell BCV, Majoie CBLM, Albers GW, Menon BK, Yassi N, Sharma G, van Zwam WH, van Oostenbrugge RJ, Demchuk AM, Guillemin F, et al; HERMES collaborators. Penumbra imaging and functional outcome in patients with anterior circulation ischaemic stroke treated with endovascular thrombectomy versus medical therapy: a meta-analysis of individual patient-level data. Lancet Neurol. 2019;18:46-55.
12. Albers GW, Marks MP, Kemp S, Christensen S, Tsai JP, Ortega-Gutierrez S, McTaggart RA, Torbey MT, Kim-Tenser M, Leslie-Mazwi T, et al. DEFUSE 3 Investigators. Thrombectomy for Stroke at 6 to 16 Hours with Selection by Perfusion Imaging. N Engl J Med. 2018 Feb 22;378(8):708-718.
13. Jovin TG, Nogueira RG, Lansberg MG, Demchuk AM, Martins SO, Mocco J, Ribo M, Jadhav AP, Ortega-Gutierrez S, Hill MD et al. Thrombectomy for anterior circulation stroke beyond 6 h from time last known well (AURORA): a systematic review and individual patient data meta-analysis. Lancet. 2022 Jan 15;399(10321):249-258.
14. Pujara D, Nogueira RG, Saver JL, Li L, Han Q, Dai Y, Sang H, Yang Q, Nguyen TN, Qiu Z. Mechanical Thrombectomy for Large Ischemic Stroke: A Systematic Review and Meta-analysis. Neurology. 2023 Aug 29;101(9):e922-e932.
15. Yoshimura S, Sakai N, Yamagami H, Uchida K, Beppu M, Toyoda K, Matsumaru Y, Matsumoto Y, Kimura K, Takeuchi M, et al. Endovascular Therapy for Acute Stroke with a Large Ischemic Region. N Engl J Med. 2022 Apr 7;386(14):1303-1313.

16. Huo X, Ma G, Tong X, Zhang X, Pan Y, Nguyen TN, Yuan G, Han H, Chen W, Wei M, et al. ANGEL-ASPECT Investigators. Trial of Endovascular Therapy for Acute Ischemic Stroke with Large Infarct. N Engl J Med. 2023 Apr 6;388(14):1272-1283.
17. Sarraj A, Hassan AE, Abraham MG, Ortega-Gutierrez S, Kasner SE, Hussain MS, Chen M, Blackburn S, Sitton CW, Churilov L et al. SELECT2 Investigators. Trial of Endovascular Thrombectomy for Large Ischemic Strokes. N Engl J Med. 2023 Apr 6;388(14):1259-1271.
18. Bendszus M, Fiehler J, Subtil F, Bonekamp S, Aamodt AH, Fuentes B, Gizewski ER, Hill MD, Krajina A, Pierot L et al. TENSION Investigators. Endovascular thrombectomy for acute ischaemic stroke with established large infarct: multicentre, open-label, randomised trial. Lancet. 2023 Nov 11;402(10414):1753-1763
19. Costalat V, Jovin TG, Albucher JF, Cognard C, Henon H, Nouri N, Gory B, Richard S, Marnat G, Sibon I et al. LASTE Trial Investigators. Trial of Thrombectomy for Stroke with a Large Infarct of Unrestricted Size. N Engl J Med. 2024 May 9;390(18):1677-1689.
20. Uchida K, Shindo S, Yoshimura S, Toyoda K, Sakai N, Yamagami H, Matsumaru Y, Matsumoto Y, Kimura K, Ishikura R et al. RESCUE-Japan LIMIT Investigators. Association Between Alberta Stroke Program Early Computed Tomography Score and Efficacy and Safety Outcomes With Endovascular Therapy in Patients With Stroke From Large-Vessel Occlusion: A Secondary Analysis of the Recovery by Endovascular Salvage for Cerebral Ultra-acute Embolism-Japan Large Ischemic Core Trial (RESCUE-Japan LIMIT). JAMA Neurol. 2022 Dec 1;79(12):1260-1266
21. Yoo AJ, Zaidat OO, Sheth SA, Rai AT, Ortega-Gutierrez S, Given CA 2nd, Zaidi SF, Grandhi R, Cuellar H, Mokin M et al. TESLA Investigators. Thrombectomy for Stroke With Large Infarct on Noncontrast CT: The TESLA Randomized Clinical Trial. JAMA. 2024 Sep 23;332(16):1355–66.
22. Khalife J, Tonetti DA, Jovin TG. CT-Guided Thrombectomy for Large Core Stroke Up to 24 Hours-Another Piece in a Complex Puzzle. JAMA. 2024 Sep 23.
23. Chen H, Lee JS, Michel P, Yan B, Chaturvedi S. Endovascular Stroke Thrombectomy for Patients With Large Ischemic Core: A Review. JAMA Neurol. 2024 Oct 1;81(10):1085-1093.
24. Baker WB, Sun Z, Hiraki T, Putt ME, Durduran T, Reivich M, Yodh AG, Greenberg JH. Neurovascular coupling varies with level of global cerebral ischemia in a rat model. J Cereb Blood Flow Metab. 2013;33:97-105. doi: 10.1038/jcbfm.2012.137.
25. Florence G, Guerit JM, Gueguen B. Electroencephalography (EEG) and somatosensory evoked potentials (SEP) to prevent cerebral ischaemia in the operating room. Neurophysiol Clin. 2004;34:17-32.
26. Martinez - Piñeiro A, Lucente G, Hernandez-Perez M, Cortés J, Arbex A, Perez de la Ossa N, Ramos-Fransi A, et al. Prognostic Accuracy of N20 Somatosensory Potential in Patients With Acute Ischemic Stroke and Endovascular Thrombectomy. Stroke: Vascular and Interventional Neurology 2024 <https://doi.org/10.1161/SVIN.122.000735>

 timeisbrain	CLINICAL INVESTIGATION PLAN PROMISE20 RELOADED	CODE: PROMISE20 RELOADED CIP REV: 1 DATE: 11/04/2025 PAGE: 44 of 44
--	---	---

22. ANNEXES

1. REG.7.3-03-11 Investigational Brochure – IB.
2. REG.7.3-03-07 HIP+CI v4.0.
3. REG.7.3-03-08 AE Adverse event.
4. REG.7.3-03-09 DD Device deficiency form.
5. REG.7.3-03-10 Case report form-CRF.
6. OP.8.2-03 Vigilance system.
7. Clinical evaluation 2025.
8. Post-market clinical follow up plan-PMCFP 2025.

PROMISE20 RELOADED CIP v1

Final Audit Report

2025-04-10

Created:	2025-04-10
By:	TIB DOCONTROL (timeisbrainraqa@gmail.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAH-pGEPWQxoNOvzfEcjpls5wsAFmlFBmA

"PROMISE20 RELOADED CIP v1" History

-  Document created by TIB DOCONTROL (timeisbrainraqa@gmail.com)
2025-04-10 - 11:37:30 AM GMT
-  Document emailed to Antoni Dávalos (adavalos.germanstrias@gmail.com) for signature
2025-04-10 - 11:37:37 AM GMT
-  Email viewed by Antoni Dávalos (adavalos.germanstrias@gmail.com)
2025-04-10 - 3:01:29 PM GMT
-  Document e-signed by Antoni Dávalos (adavalos.germanstrias@gmail.com)
Signature Date: 2025-04-10 - 3:01:47 PM GMT - Time Source: server
-  Agreement completed.
2025-04-10 - 3:01:47 PM GMT